

Protective Role of *Syzygium Cymosum* Leaf Extract Against Carbofuran-Induced Hematological and Hepatic Toxicities

Tania Binte Wahed,[†] Milon Mondal,[‡] Mohammad Asikur Rahman,[†] Md. Sakib Hossen,[§] Nikhil Chandra Bhoomik,[¶] Sushmita Saha,[†] E. M. Tanvir,^{||} Md. Ibrahim Khalil,[⊥] Sukalyan Kumar Kundu,[†] Muhammad Torequl Islam,^{*,∇,●} and Mohammad S. Mubarak^{*,#,▲,ID}

[†]Department of Pharmacy, Jahangirnagar University, Savar, Dhaka 1342, Bangladesh

[‡]Department of Pharmacy, Bangabandhu Sheikh Mujibur Rahman Science and Technology University, Gopalganj 8100, Bangladesh

[§]Department of Biochemistry, Primeasia University, Banani 1213, Bangladesh

[¶]Wazed Miah Science Research Centre, Jahangirnagar University, Savar, Dhaka 1342, Bangladesh

^{||}Institute of Food and Radiation Biology, Atomic Energy Research Establishment, Savar, Bangladesh

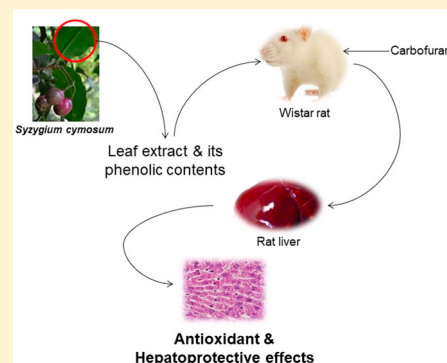
[⊥]Department of Biochemistry and Molecular Biology, Jahangirnagar University, Savar, Dhaka 1342, Bangladesh

[∇]Department for Management of Science and Technology Development, Ton Duc Thang University, Ho Chi Minh City 700000, Vietnam

[●]Faculty of Pharmacy, Ton Duc Thang University, Ho Chi Minh City 700000, Vietnam

[#]Department of Chemistry, The University of Jordan, Amman 11942, Jordan

ABSTRACT: The aim of the present study was to evaluate the protective effect of *Syzygium cymosum* leaf methanol extract (SCL) against carbofuran (CF)-induced hepatotoxicity in Sprague–Dawley rats, along with the identification and quantification of polyphenolic composition by high-performance liquid chromatography (HPLC). Results revealed the presence of alkaloids, tannins, and flavonoids in SCL. Similarly, HPLC analysis suggests that SCL contains some known important antioxidants, such as rutin, benzoic acid, and salicylic acid that could be responsible for the hepatoprotective activity of the extract. In CF-exposed rats, significant hematological alterations along with histological changes were marked by the presence of necrosis, congestion, and inflammation. CF-intoxication also showed an increase in lipid peroxidation and decrease in cellular antioxidant enzymes (e.g., superoxide dismutase, catalase, and glutathione peroxidase) levels in rats compared with the control group. Furthermore, coadministration of SCL significantly ameliorated the abnormalities and improved the cellular arrangement in experimental animals. SCL also reversed the alteration of hematological and biochemical parameters and brought them back to normal levels as compared to the control group. In conclusion, *S. cymosum* may be one of the best sources of natural antioxidant compounds that can be used in the treatment of oxidative stress and stress-related diseases and disorders.



INTRODUCTION

Carbofuran (CF), chemically known as 2,3-dihydro-2,2-dimethyl-7-benzofuranyl *N*-methylcarbamate, is a widely used pesticide applied for agricultural development.¹ Continuous exposure to different kinds of toxic pollutants and pathogens in modern civilization may cause severe acute and chronic health disorders.² Along this line, most xenobiotics are metabolized in the liver, and during metabolism, CF generates deleterious free radicals that may trigger oxidative stress, which in turn causes liver cirrhosis, acute, and chronic diseases.³ Therefore, natural antioxidant therapy received higher recommendation for patients who are suffering from hepatic diseases.⁴ Antioxidants of natural sources are the most powerful reducing agents to diminish oxidative tissue injuries as well as prevent noxious effects. Although there are massive advances in the modern

medicine system, reliable drugs for liver function protection and stimulation for hepatic cell regeneration are few in number.⁵ In this context, synthetic antioxidants may have adverse side effects, almost zero nutritional qualities, and great effects on metabolism.⁶ Hence, antioxidants from natural sources are being given priority in seeking protection against oxidative stress and related diseases and disorders.⁷ Moreover, herbs, herbal extracts, or phytochemicals are currently broadly used as foods, drugs, and as traditional medicines.⁸ The global economy of the international trade of herbal products has been increasing by 15% annually, with raw material for most herbal products being sourced from South and Southeast Asian

Received: April 16, 2019

Published: July 23, 2019



Table 1. Phytochemical Constituents Identified in the Methanol Leaf Extract of *S. cymosum*^a

phytochemicals	name of the test	observed changes	result
alkaloids	Mayer's test	creamy white precipitate	+
	Hager's test	yellow crystalline precipitate	+
	Wagner's test	brown or deep brown precipitate	++
	Dragendorff's test	orange or orange-red precipitate	++
carbohydrates	Molisch's test	a red or reddish violet ring is formed at the junction of two layer, and on shaking a dark purple solution is formed	++
glycosides	general test	yellow color	—
	test for glucoside	production of brick-red precipitation (carried out with the hydrolyzed extract)	—
flavonoids	general test	red color	++
saponins	frothing test	formation of stable foam	—
steroids	Liebermann–Burchard's test	greenish color	—
tannins	lead acetate test	a yellow or red precipitate	++

^a[+ = Presence, ++ = Strong presence, — = Absence]; SCL= *S. cymosum* leaf extract.

countries.⁹ Because of the popularity of these herbal products, in some countries, they were divided into two categories—herbal medicine (HM) and traditional herbal product (THP)—to accommodate new legislation for herbal product licensing.¹⁰ Therefore, herbal products may be one of the potential sources of therapeutic agents beneficial for designing future drugs.¹¹

Syzygium cymosum is an evergreen tree growing up to 20 m tall, and the bole can be 50 cm in diameter, with edible fruits.¹² Dhar and co-workers have recently investigated the methanolic bark extract of *S. cymosum*. These researchers found that the bark contained phytoconstituents including carbohydrates, alkaloid, glycosides, steroids, tannins, and saponins and showed significant antioxidant and cytotoxic activities.¹³ To the best of our knowledge, no studies have been done on the antioxidant and hepatoprotective activity of SCL. Accordingly, the present study was designed to investigate the antioxidant property of the methanolic leaf extract of *S. cymosum* and to correlate the antioxidant capacity with hepatoprotective activity against CF-induced hepatotoxicity.

RESULTS

Phytochemical Screening. Preliminary phytochemical screening of the crude leaf extract revealed that *S. cymosum* contains different types of compounds, in which alkaloids, tannins, and flavonoids were predominant. Listed in Table 1 are phytochemicals identified in the methanol *S. cymosum* leaves extract.

HPLC Analysis of Phenolic Compounds. Data obtained from HPLC analysis confirmed the presence of benzoic acid, salicylic acid, and rutin in SCL (Figure 1). Phenolic and flavonoid compounds quantified in the methanol extract of *S. cymosum* are presented in Table 2.

Total Polyphenol, Total Flavonoid, and Total Antioxidant Capacity. Phenolics are vital phytoconstituents that act as singlet oxygen quenchers and free radical scavengers to minimize molecular damage. In this study, our results show that the total polyphenol content of the methanolic leaf extract of *S. cymosum* was 21.70 ± 0.42 g GAE/100 g of SCL, whereas the total flavonoid content was lower (6.45 ± 0.53 g QE/100 g of SCL), which is often the case for most plants. However, the total antioxidant capacity of *S. cymosum* leaf extract was 20.25 ± 0.28 mg AAE/100 g of SCL.

DPPH and NO Radical Scavenging. In this investigation, the DPPH and NO assays were conducted to evaluate the

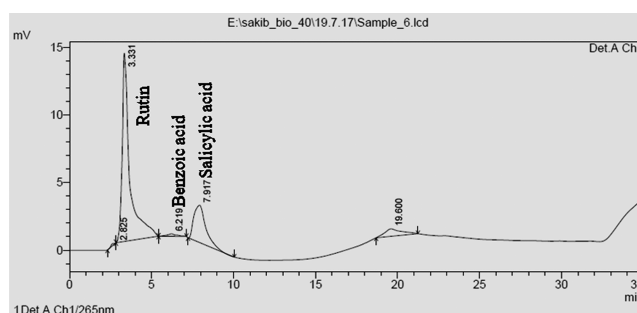


Figure 1. HPLC chromatograph of *S. cymosum* leaf extract.

antioxidant activity of SCL. Results revealed that the extract exhibits significant free-radical-scavenging capacity when compared to ascorbic acid, used as a standard. In the DPPH assay, IC₅₀ values calculated for SCL and ascorbic acid were 122.11 and 25.53 μ g/mL, respectively (Figure 2a). On the other hand, the IC₅₀ values calculated for the leaf extract and ascorbic acid were 109.46 and 45.23 μ g/mL, respectively (Figure 2b).

Reducing Power Assessment. Reducing power of SCL as an antioxidant was assessed by the ferric and cupric assay. Our findings suggest that SCL exhibits a moderate concentration-dependent ferric and cupric reducing power as compared to ascorbic acid (Figure 3a, 3b).

Acute Toxicity Study. Administration of up to 4000 mg/kg of SCL did not cause any mortality in animals (Swiss albino rats). In addition, animals did not show any sign of restlessness, respiratory distress, general irritation, coma, or convulsion. Hence, this extract could be considered safe for animal consumption.

Effects on Body Weight and Relative Liver Weight. Shown in Table 3 are results of our study of the effects of the methanol extract of *S. cymosum* leaves and CF on the body and organ weights in different experimental groups. Our findings reveal that treatment of animals with CF alone caused a significant weight loss, in addition to an increase in the relative liver weight. However, coadministration with SCL and with the standard (Silymarin) ameliorated the body weight and relative liver weight close to the control group.

Effects on Food and Water Intake. Animals administered with CF alone showed less consumption of food and water. However, animals treated with the extract (SCL 250 + CF) responded almost the same as those treated with

Table 2. Phenolic and Flavonoid Compounds Detected in *S. cymosum* Leaf Extract Using HPLC Analysis^a

standard compounds	retention time of standard (min)	retention time of SCL (min)	area	concentration (mg/g)
benzoic acid	6.227	6.219	7860	0.045 ± 0.0015*
catechin	3.809	-	-	ND
gallic acid	3.696	-	-	ND
naringin	5.095	-	-	ND
pyrogallol	5.212	-	-	ND
quercetin	19.263	-	-	ND
rutin	3.449	3.331	448495	0.576 ± 0.002 [#]
salicylic acid	8.660	7.917	138217	2.574 ± 0.017 [@]
tannic acid	3.178	-	-	ND
trans-cinnamic acid	16.722	-	-	ND
vanillic acid	19.623	-	-	ND

^aData are presented as the mean ± SEM, (*n* = 3); values in the same column with different superscripts are significantly different at *p* < 0.05. ND: Not detected. SCL: *S. cymosum* leaf extract.

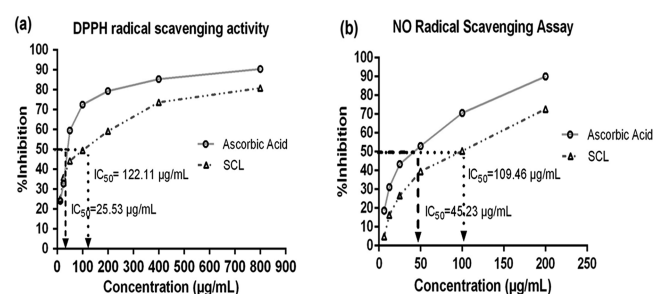


Figure 2. (a) DPPH-free-radical-scavenging activity and (b) NO-radical-scavenging activity of SCL and ascorbic acid. SCL: *S. cymosum* leaf extract. IC₅₀: Inhibitory concentration 50%.

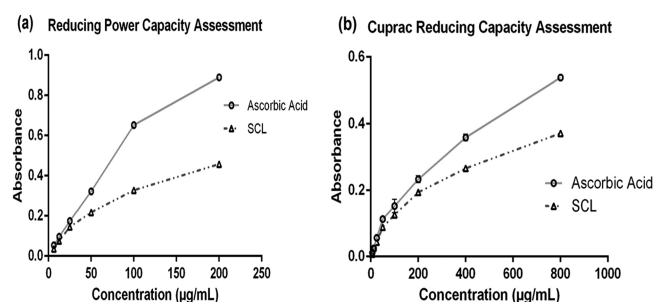


Figure 3. (a) Ferric reducing power capacity and (b) cupric reducing capacity of *S. cymosum*.

silymarin. Moreover, rats treated with SCL at 500 mg/kg BW (SCL 500 + CF) showed a better response to food and water intake than those treated with the standard drug (silymarin) as depicted in Figure 4a,b.

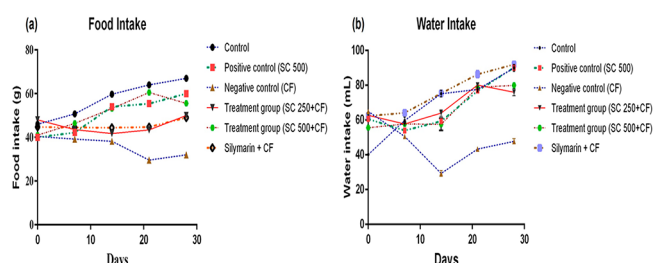


Figure 4. Effects of different treatment groups on (a) food intake and (b) water intake.

Effects on Hematological Parameters. Our results from this study showed that the red blood cells (RBC) and mean corpuscular volume (MCV) levels significantly increased in rats when SCL 500 was administered alone. On the contrary, CF administration significantly decreased RBC, hemoglobin (HGB), and hematocrit (HCT) levels but increased MCV and MCH levels. Similarly, treatment groups (SC 250 + CF) and (SC 500 + CF) displayed a significant increase in HGB and HCT levels, and a decrease in MCV and MCH levels when compared with rats treated with CF alone as shown in Table 4. On the other hand, results revealed that the levels of WBC, lymphocyte and monocyte, increased when rats were administered with CF alone, which was ameliorated by treatment with SCL at 250 and 500 mg/kg doses [(SCL 250 + CF) and (SCL 500 + CF)] groups (Table 4). In the case of platelet count, CF treatment caused a significant decrease in platelet count; however, mean platelet volume (MPV) and plateletcrit (PCT) were found to increase significantly as compared with the control group. Finally, administration of SCL ameliorated these parameters in the treated groups [(SCL 250 + CF) and (SCL 500 + CF)] (Table 4).

Table 3. Effect of the Methanol Extract of *S. cymosum* Leaves and of Carbofuran on Body Weight and Relative Organ Weight in Rats^a

groups	initial BW (g)	final BW (g)	BW gain (g)	relative liver weight (g)
control	156.00 ± 3.91	196.60 ± 5.43	40.60 ± 3.14	4.25 ± 0.01
positive control (SCL 500)	154.40 ± 16.72	191.20 ± 14.05	36.80 ± 3.34	4.28 ± 0.07
negative control (CF)	154.20 ± 3.01	178.00 ± 4.90	23.80 ± 4.08*	4.93 ± 0.01**
treatment group (SCL 250 + CF)	155.20 ± 12.42	183.00 ± 9.78	27.80 ± 3.85	4.34 ± 0.04*
treatment group (SCL 500 + CF)	156.20 ± 10.91	186.60 ± 10.77	30.40 ± 1.03	4.28 ± 0.03
silymarin + CF	158.40 ± 15.87	189.60 ± 15.35	31.20 ± 1.98	4.27 ± 0.02

^aValues represent the mean of 7 rats ± SEM. Significant at **p* > 0.05 and ***p* > 0.001, as compared with the corresponding control group.

Table 4. Effects of the Methanol Extract of *S. cymosum* Leaves and of Carbofuran on Hematological Parameters^a

	hematological parameters	groups					
		control	positive control (SCL 500)	negative control (CF)	treatment group (SCL 250+ CF)	treatment group (SCL 500+ CF)	silymarin + CF
erythrocyte count	RBC ($10^6/\mu\text{L}$)	7.65 $\pm$ 0.02	8.07 $\pm$ 0.09*	6.78 $\pm$ 0.01**	7.86 $\pm$ 0.03	7.70 $\pm$ 0.09	7.54 $\pm$ 0.07
	HGB (g/dL)	13.41 $\pm$ 0.01	13.33 $\pm$ 0.05	8.81 $\pm$ 0.04**	13.18 $\pm$ 0.03*	13.56 $\pm$ 0.04	13.33 $\pm$ 0.04
	HCT (%)	51.67 $\pm$ 0.15	51.87 $\pm$ 0.01	49.79 $\pm$ 0.09**	51.14 $\pm$ 0.15*	51.48 $\pm$ 0.02	51.64 $\pm$ 0.03
	MCV (fL)	53.39 $\pm$ 0.30	56.62 $\pm$ 0.12**	62.94 $\pm$ 0.04**	50.32 $\pm$ 0.06**	58.36 $\pm$ 0.19**	52.71 $\pm$ 0.04*
	MCH (pg)	16.38 $\pm$ 0.03	16.31 $\pm$ 0.01	16.49 $\pm$ 0.02*	16.48 $\pm$ 0.07*	16.39 $\pm$ 0.04	16.37 $\pm$ 0.06
	RDW-CV (%)	25.77 $\pm$ 0.29	25.78 $\pm$ 0.26	26.44 $\pm$ 0.36	26.19 $\pm$ 0.80	25.88 $\pm$ 0.40	26.14 $\pm$ 0.26
leukocyte count	WBC	4.30 $\pm$ 1.03	4.37 $\pm$ 0.28	7.33 $\pm$ 0.50**	6.27 $\pm$ 0.59**	5.52 $\pm$ 0.52**	4.51 $\pm$ 0.48*
	lymphocyte (%)	70.26 $\pm$ 2.40	72.69 $\pm$ 0.03**	83.12 $\pm$ 0.90**	79.97 $\pm$ 0.78**	76.29 $\pm$ 0.26**	73.47 $\pm$ 0.26**
	monocyte (%)	3.41 $\pm$ 0.80	3.42 $\pm$ 0.17	3.49 $\pm$ 0.20*	3.51 $\pm$ 0.57**	3.48 $\pm$ 0.12*	3.41 $\pm$ 0.14
	neutrophil (%)	31.24 $\pm$ 0.93	32.61 $\pm$ 0.62	28.22 $\pm$ 1.31	29.80 $\pm$ 0.49	31.52 $\pm$ 0.49	31.11 $\pm$ 0.69
platelet count	platelet ($10^3/\mu\text{L}$)	712.33 $\pm$ 8.50	769.67 $\pm$ 2.60**	433.67 $\pm$ 6.65**	515.33 $\pm$ 3.84**	633.67 $\pm$ 6.98**	704.00 $\pm$ 7.54
	MPV (fL)	7.72 $\pm$ 0.62	7.67 $\pm$ 0.35	7.23 $\pm$ 0.75**	7.32 $\pm$ 0.63**	7.72 $\pm$ 0.39	7.69 $\pm$ 0.55
	PDW (fL)	8.57 $\pm$ 0.25	8.48 $\pm$ 0.26	9.17 $\pm$ 0.12	9.13 $\pm$ 0.12	9.15 $\pm$ 0.25	8.31 $\pm$ 0.18
	PCT (%)	0.56 $\pm$ 0.11	0.61 $\pm$ 0.24	0.71 $\pm$ 0.30*	0.58 $\pm$ 0.33	0.58 $\pm$ 0.26	0.57 $\pm$ 0.19
	P-LCR (%)	9.79 $\pm$ 0.72	9.72 $\pm$ 0.28	8.47 $\pm$ 0.36	7.60 $\pm$ 0.36	10.20 $\pm$ 0.61	9.55 $\pm$ 0.71

^aValues are mean $\pm$ SEM ($n = 7$). Significant at * $p > 0.05$ and ** $p > 0.001$, as compared with the control group.

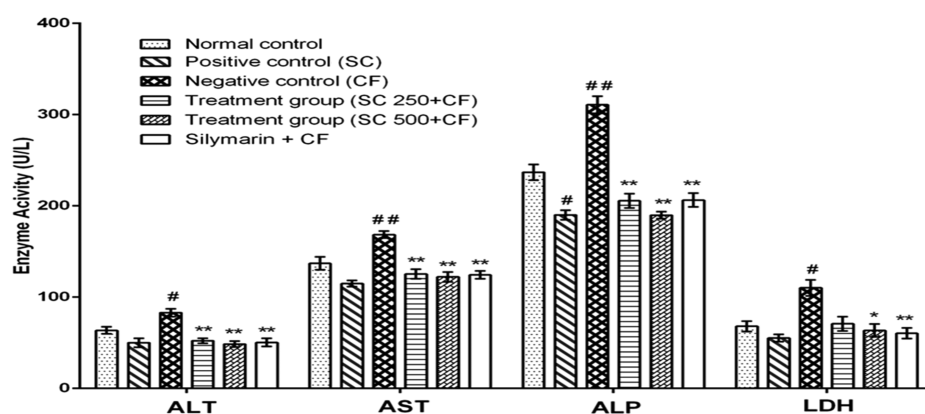


Figure 5. Effects of carbofuran and *S. cymosum* leaf extracts on serum hepatocellular function biomarkers. Values are expressed as the mean $\pm$ SEM ($n = 7$). Significant at p values: # < 0.05 , ## < 0.001 compared with the control group. p values: * < 0.05 , ** < 0.001 compared with negative control (CF) group.

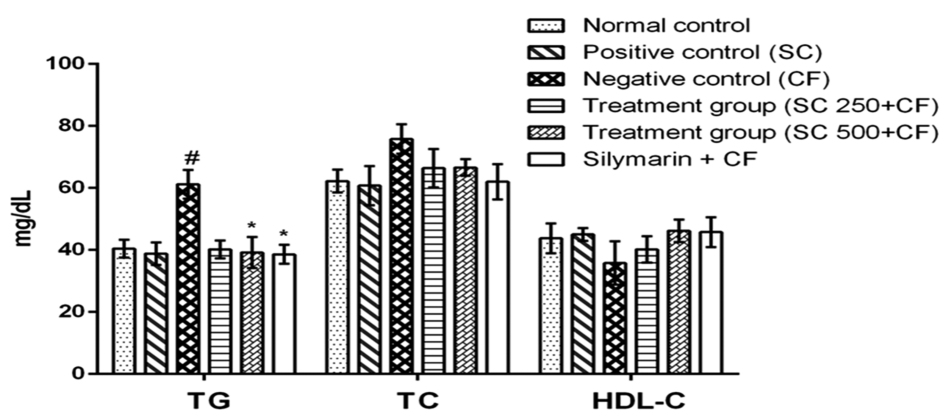


Figure 6. Effects of carbofuran and *S. cymosum* leaf extracts on serum lipid profile. Values are expressed as the mean $\pm$ SEM ($n = 7$). Significant at p values: # < 0.05 , ## < 0.001 compared with the control group. p values: * < 0.05 , ** < 0.001 compared with negative control (CF) group.

Effect of the Methanol Extract of *S. cymosum* Leaves on Biochemical Parameters. Our findings from this investigation showed that the serum activities of alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase (ALP), and lactate dehydrogenase (LDH) enzymes significantly increased in the CF group in

comparison with the control group. However, coadministration of SCL and silymarin lowered the levels of these serum enzymes (Figure 5). On the other hand, the serum triglycerides (TG) levels were markedly augmented in CF-intoxicated rats, but coadministration of SCL with CF attenuated the elevated TG level. Silymarin displayed a similar

effect on these parameters. Furthermore, serum total cholesterol (TC) level increased and high density lipoprotein cholesterol (HDL) level decreased in a nonsignificant manner when rats were administered with CF alone compared with the control group (Figure 6).

Similarly, the serum total bilirubin level was higher in the rats administrated with the CF group when compared to control group animals. However, treatment with SCL at 500 mg/kg dose caused significant amelioration in serum total bilirubin level compared with the silymarin treatment group (Figure 7). No significant difference was observed between

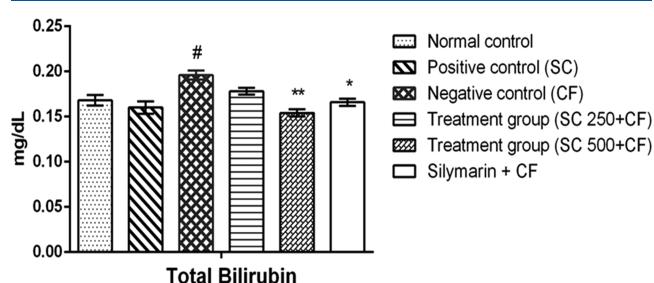


Figure 7. Effects of carbofuran and *S. cymosum* leaf extracts on serum total bilirubin level. Values are expressed as mean $\pm$ SEM ($n = 7$). Significant at p values: $\# < 0.05$, $\#\# < 0.001$ compared with the control group. p values: $* < 0.05$, $** < 0.001$ compared with the negative control (CF) group.

control and CF groups in serum total protein level. On the other hand, SCL and silymarin groups showed no significant difference in serum albumin level compared to the control group. Taken all together, a significant improvement was observed in both SCL (SCL 500 + CF) as well as in the silymarin groups. Moreover, a slight difference was found in the serum albumin level between SCL and silymarin-treated groups (Table 5).

Table 5. Effects of Carbofuran and *S. cymosum* Leaf Extracts on Serum Total Protein and Albumin Levels

groups	total protein (g/dL)	albumin (g/dL)
normal control	6.94 $\pm$ 0.06	3.00 $\pm$ 0.07
positive control (SCL 500)	6.86 $\pm$ 0.04	2.98 $\pm$ 0.07
negative control (CF)	6.78 $\pm$ 0.21	2.54 $\pm$ 0.05 [#]
treatment group (SCL 250 + CF)	6.94 $\pm$ 0.07	2.75 $\pm$ 0.04
treatment group (SCL 500 + CF)	6.96 $\pm$ 0.05	2.96 $\pm$ 0.05 ^{**}
silymarin + CF	6.92 $\pm$ 0.06	2.98 $\pm$ 0.04 ^{**}

^aValues are expressed as the mean $\pm$ SEM ($n = 7$). Significant at p values: $\# < 0.05$, $\#\# < 0.001$ compared with the control group. p values: $* < 0.05$, $** < 0.001$ compared with the negative control (CF) group.

Effect of the Methanol Extract of *S. cymosum* Leaves on Lipid Peroxidation. Our results of the effects of CF and SCL on the levels of lipid peroxidation in the liver tissue of control and experimental rats are illustrated in Figure 8. MDA levels were significantly ($p < 0.001$) higher in CF-intoxicated rats when compared with the control group. However, treatment with SCL followed by CF administration ameliorated MDA level significantly ($p < 0.001$), conferring the hepatocellular protection of SCL against CF-induced elevation of lipid peroxidation as evidenced by a significant decrease in MDA levels.

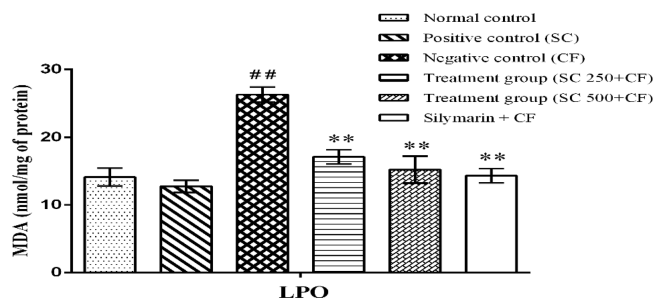


Figure 8. Effects of carbofuran and *S. cymosum* leaf extract on LPO. Values are expressed as the mean $\pm$ SEM ($n = 7$). Significant at p values: $\# < 0.05$, $\#\# < 0.001$ compared with the control group. p values: $* < 0.05$, $** < 0.001$ compared with the negative control (CF) group.

Effects of the Methanol Extract of *S. cymosum* Leaves on Hepatic Antioxidant Enzymes.

Presented in Table 6 are results related to activities of the cellular antioxidant enzymes, superoxide dismutase (SOD), catalase (CAT), and glutathione peroxidase (GPx) in the liver of control and experimental rats. Results indicate a significant ($p < 0.001$) decrease in the activities of these antioxidant enzymes in rats treated with CF alone when compared with the control group. However, oral coadministration of SCL followed by CF administration for 28 days significantly ($p < 0.001$) restored the activities of these enzymes to their near normal levels as compared to the CF-treated group.

Histological Examination of Liver Tissue. Microscopic evaluation of the liver tissues indicates that animals from the normal control and positive control (SC) groups had a regular arrangement of hepatocytes scattering from the central vein to the periphery of the lobule. Nuclei of the hepatocytes also showed normal vesicular structure with a general and uniform cytoplasm observed as depicted in Figure 9a,b. In contrast, the CF-treated group showed severe disruption of the cellular arrangement, vascular congestion, degeneration of hepatocytes at the peripheral area of the central vein (CV) associated with inflammatory infiltrates, necrosis of the hepatocytes, and edema at different locations of the lobule (Figure 9c). These histopathological abnormalities induced by CF were ameliorated in the liver of rats in the treated groups in all of the investigated histopathological features. A moderate to mild degree of worsening of hepatocytes and a mild congestion of CV with inflammation were also observed (Figure 9d,e). On the other hand, Figure 9f shows the standard silymarin group with marked preservation in the cellular arrangement. These consequences of histopathologic semiquantitative scorings are presented in Table 7.

DISCUSSION

To the best of our knowledge, this is the first experimental study pertaining to the protective effects of SCL in CF-induced toxicity in experimental animals. Bioactive compounds in SCL were identified by HPLC analysis. Results showed that alkaloids, carbohydrates, flavonoids, and tannins are the major phytochemicals found in SCL. These chemical constituents are known to be biologically active and are considered as secondary metabolites which are responsible for antioxidant, antimicrobial, anticancer, and antifungal activities.^{14,15} In addition, plant phenolic compounds and flavonoids have hydrogen or electron donating ability to resist/scavenge/stop generation of free radicals in oxidative

Table 6. Effects of Carbofuran and *S. cymosum* Leaf Extracts on the Tissue Activities of Antioxidant Enzymes in the Rat Liver^a

groups	SOD (units/mg tissue protein)	CAT (units/mg tissue protein)	GPx (mIU/mL)
normal control	4.49 ± 0.10	27.74 ± 1.06	4.24 ± 0.02
positive control (SCL 500)	4.14 ± 0.09	25.50 ± 0.62	4.54 ± 0.11
negative control (CF)	1.61 ± 0.11 [#]	13.86 ± 0.80 [#]	1.94 ± 0.02 [#]
treatment group (SCL 250 + CF)	2.93 ± 0.09 ^{**}	20.04 ± 0.67 ^{**}	2.77 ± 0.12 ^{**}
treatment group (SCL 500 + CF)	3.96 ± 0.05 ^{**}	23.74 ± 0.55 ^{**}	4.01 ± 0.06 ^{**}
silymarin + CF	4.04 ± 0.12 ^{**}	25.18 ± 0.86 ^{**}	4.18 ± 0.07 ^{**}

^aValues are the mean ± SEM ($n = 7$). Significant at p values: [#] < 0.05, [#] < 0.001 compared with the control group. P values: * < 0.05, ** < 0.001 compared with the negative control (CF) group.

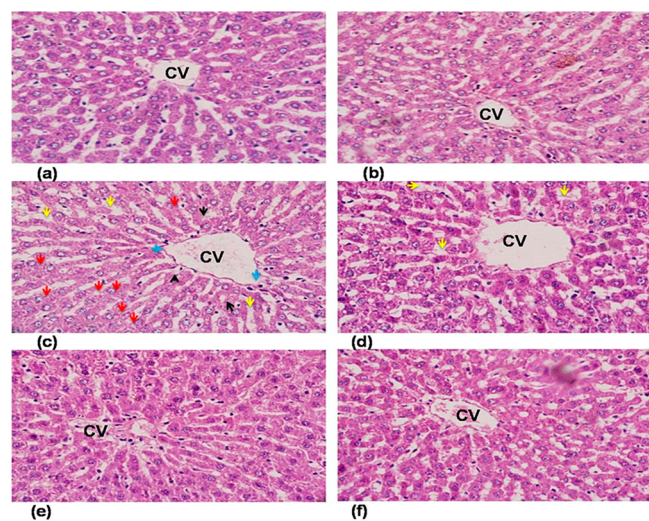


Figure 9. Liver sections from experimental animals in (a) normal control group showing a normal liver with a hepatic lobule and a uniform pattern of polyhedral hepatocytes radiating from the central vein (CV) toward the periphery, (b) positive control rats revealing a normal appearance of hepatocytes surrounding the CV, and (c) negative control rats showing severe disruption of the cellular arrangement radiating from the central vein (CV) and in the lobule. There was marked necrosis of hepatocytes in the peripheral area of the CV (black arrows) and in the lobule (red arrows). In addition, congestion in the CV associated with inflammatory infiltrates (blue arrow) and edema at different locations (yellow arrow) in the lobule was observed. (d and e) liver section of the animals treated with SCL 250 mg/kg + CF and SCL 500 mg/kg + CF, respectively, showing a remarkable degree of preservation in the cellular arrangement with only mild inflammation. (f) Silymarin + CF group also showing marked preservation in the cellular arrangement [magnification: 40 $\times$; scale bar: 20 μ m].

stress.¹⁶ To combat harmful oxidative damage, dietary antioxidant supplementations may be regarded as the defense mechanism against highly reactive toxins.¹⁷ Results obtained

from DPPH- and NO-scavenging assays, and from ferric and cupric reducing capacity, validated the high antioxidant potential of SCL.

In addition, HPLC analysis confirmed the presence of benzoic acid, rutin, and salicylic acid in SCL. These bioactive compounds provide a good indication for antioxidant activity due to their redox potentials, which play a vital role in neutralizing free radicals, scavenging reactive oxygen species (ROS), and degrading peroxides.¹⁸ In this context, benzoic acid is an aromatic carboxylic acid which occurs naturally in many plants and is used as a food preservative. Benzoic acid and its derivatives are known to possess potent anti-inflammatory, antipyretic, and antioxidant properties.¹⁹ Saravanan and Nalini reported that 2-hydroxy-4-methoxy benzoic acid exerts hepatoprotective ability against ethanol-induced toxicity in experimental animals.²⁰ On the other hand, rutin is the natural glycoside combining the flavonol quercetin and the disaccharide rutinose (α -L-rhamnopyranosyl-(1 $\rightarrow$ 6)- β -D-glucopyranose). It is a citrus flavonoid found in numerous plants, including citrus fruit. This compound generally binds to Fe²⁺, thereby preventing it from binding to hydrogen peroxide, and thus inhibits free radical-induced cellular damage.²¹ Presence of the rutinose moiety is crucial for some of the protective effects of rutin.²² Similarly, salicylic acid is a phenolic compound with antioxidant properties and is capable of scavenging free radicals. It additionally exhibits antiaging properties as well as reducing the risk of cancer.²³ Zhao and colleagues showed that salicylic acid exerts protective effect against sulfur dioxide-induced lipid peroxidation in mice.²⁴ Recently,^{25–27} research findings indicated that salicylic acid counteracts oxidative damage, which was induced by adverse conditions in animals, and exhibited protective effects against paclitaxel and cisplatin-induced neurotoxicity.²⁸

In organ toxicity studies, body weight and relative organ weights are important criteria for evaluation of the organ damage.^{29,30} In this respect, significant changes in the rats' body weights, along with relative organ weights, may provide an imperative indication about CF-induced toxicity. Our study

Table 7. Semi-Quantitative Scoring of the Architectural Changes in the Histopathological Examination of Rat Liver^a

scoring parameters	group					
	normal control	positive control (SCL 500)	negative control (CF)	treatment group (SCL 250 + CF)	treatment group (SCL 500 + CF)	silymarin + CF
degeneration of hepatocytes	—	—	+++	++	+	+
inflammatory cell infiltration	—	—	+++	++	+	+
vascular congestion	—	—	+++	++	+	+
edema	—	—	+++	+++	+	+

^aScoring was expressed as follows: none (—), mild (+), moderate (++), and severe (+++).

revealed that oral administration of CF nonsignificantly decreases the rats' body weights in the negative control (CF) group compared with the normal control group. However, rats treated with SCL recovered the body weight nonsignificantly close to the control group. The underlying mechanism in body weight reduction may be due to direct cytotoxic effects of the CF on somatic cells—and/or indirectly through the central nervous system which plays a role in controlling feed and water intake—and regulates the endocrine system, resulting in decreased appetite and absorption of nutrients from the gastrointestinal tract.¹ In our study, animals administrated with CF alone had less food and water intake compared with normal control rats. However, cotreatment with SCL reduced the effect of CF on food and water consumption. Nirwane and Bapat³¹ reported that toxic substances reduce food and water consumption which is in agreement with our findings. Another underlying mechanism behind the weight loss may be the rapid destruction of the cell membrane through the products of lipid peroxidation (LPO), formed by the free radical-mediated attack on membrane's phospholipids. LPO propagates an autocatalytic chain of reactions in the presence of molecular oxygen.³² Co-treatment with CF and SCL protected against cytotoxic and oxidative stress to some extent, which might have occurred because of the antioxidant potential of the extract. On the other hand, the slight increase in relative liver weight may be due to water accumulation, termed as edema. Cotreatment with SCL, however, slightly decreased the relative liver weight to near normal, which is indicative of its protective effect against the CF-induced toxicity.

Information about the internal environment of animals, as well as the pathological and physiological status of the body, can be investigated through hematological parameters and biochemical profiles.³³ The decreased levels of hematological parameters might result from the disruptive action of organophosphorus pesticides on the membranes and cell viability.³⁴ Because of pesticide action on the erythropoietin tissue, lysing or shrinkage of erythrocytes may cause a reduction in HGB and HCT levels. Exposure to CF may result in microcytic hypochromic anemia as it markedly decreases RBC, HGB, and HCT levels and increases MCV and MCH levels in rats.^{1,35} Furthermore, depletion of RBC may be attributed to hyperactivity of the bone marrow, leading to production of RBC with immaturity and with increased MCV and MCH levels. These cells can be easily destroyed in the circulation and are known as atrophied erythrocytes.^{36,37} Results from this investigation indicated that treatment of rats with SCL ameliorates levels of all investigated markers and can potentially prevent a fatal outcome. The significant increase in total WBC, monocytes, and lymphocytes levels, along with the significant decrease in the platelet count and MPV level of CF-exposed rats, indicate activation of the immune system of the rats in response of CF-induced toxicity. Findings by Celik and co-workers³⁸ demonstrated that leukocyte mobilization occurs proportional to the extent of the stress condition which may lead to leukocytosis. However, coadministration with SCL exerted significant recovery of these levels to a higher extent in CF-treated rats.

Determination of activities of serum enzymes is a crucial quantitative marker of the extent and type of hepatocellular damage. Increased levels of AST, ALT, ALP, and LDH serum have been attributed to the damaged structural integrity of the hepatic cell and the consequent release into the circulation after the autolytic breakdown or cellular necrosis in CF-treated

rats.^{39,40} These findings are consistent with that of a recent study by Hossen et al.¹ Our findings show a significant increase in the levels of ALT, AST, and ALP serum in the group coadministered with SCL. In a similar fashion, rats treated with SCL significantly restored these enzymes levels. This can be attributed to the presence of phenolic acids (such as benzoic and salicylic acid) and to flavonoids (rutin) in SCL. The inhibitory activities of these phenolic compounds with robust free-radical-scavenging abilities on membrane lipid peroxidation were also confirmed by *in vitro* assays.⁴¹

The blood lipid profile is usually investigated as an index of the degradation of hepatic cells.⁴² In this study, rats treated with SCL inhibited the damaging effects of CF by decreasing levels of TG and TC serum and by increasing HDL-C level, which is similar to the rats treated with standard drug silymarin. In addition, the serum albumin level evidently decreased in the CF-treated group, whereas coadministration of SCL with CF treatment improved albumin level. Albumin is the principal liver-made protein, and its reduction causes hepatotoxicity.⁴³ On the other hand, there was no significant difference in the serum level of total protein in SCL and the control groups. This may be ascribed to the short duration of the study.⁴⁴ In this context, Soufy and colleagues claimed that a significant increase of serum total bilirubin due to exposure to CF may arise from the vast damage of liver cells and/or blockage of the bile duct, which results in the hemolysis or gallbladder enlargement.⁴⁵ In experimental rats, exposure to CF alone increased the total bilirubin level, while treatment with either SCL or silymarin significantly reduced this parameter.

Malondialdehyde (MDA) is a toxic biomolecule generated as a byproduct from lipid peroxidation of biomembranes, and it is considered as a biomarker of oxidative stress. Increased MDA level indicates the overabundance production of free radicals resulting in oxidative hepatic damage.⁴⁶ Along this line, Rai and Sharma showed that CF administration may cause an increase in MDA level in the rat's liver and brain.⁴⁷ However, coadministration with SCL significantly reduced the levels of MDA in CF-treated rats, suggesting the protective effects of SCL against oxidative tissue damage. This protective action of SCL may be due to the presence of bioactive constituents, including benzoic acid, salicylic acid, and rutin, and their antioxidant activities to scavenge free radicals and protect the liver tissue from oxidative stress and damage.

Liver is a vital and important organ in the body and is considered a large endocrine system of the body with its own protective mechanisms through different kinds of endogenous antioxidant enzymes such as SOD, CAT, and GPx. These enzymes play significant defensive roles against oxidative stress condition⁴⁸ and abolish active oxygen species. Generally, SOD accelerates the dismutation of H₂O₂ and prevents further generation of free radicals, whereas CAT helps in the removal of H₂O₂ formed during reactions catalyzed by SOD.⁴⁹ Our results indicated that CF exposure altered SOD, CAT, and GPx activities in rat liver. Additionally, data acquired from the current investigation showed a decrease in antioxidant enzyme activity due to the increased production of ROS as evident by the increased LPO levels due to CF exposure.⁵⁰ These outcomes were in line with previous investigations which indicated that exposure to CF generates lipid peroxidation and alters the antioxidant levels of tissues in rats.¹ However, SCL supplementation elevated these antioxidative enzyme activities that had been diminished by CF. Furthermore, HPLC and *in*

vitro antioxidant results of SCL were in line with the *in vivo* results, which demonstrated that SCL can increase the activity of few antioxidant enzymes during normal physiological conditions.

Because of ROS-mediated oxidative damage, degeneration and necrosis of hepatocytes, inflammatory cell infiltrations, and vascular congestion with edematous spaces were identified in CF-intoxicated rat liver. These observations are consistent with other studies conducted on CF.¹ Oxidative hepatic damage induced by CF in rats was decreased by the coadministration of SCL, which was in good correlation with the results of serum hepatic markers, oxidative stress markers, and antioxidant enzyme activities.

■ EXPERIMENTAL SECTION

Chemicals and Reagents. Folin–Ciocalteu reagent, methanol, sodium phosphate, and ammonium molybdate were purchased from Merck, Germany. Gallic acid, ascorbic acid, quercetin, tannic acid, vanillic acid, benzoic acid, salicylic acid, pyrogallol, catechin, naringin, rutin, and 1,1-diphenyl-2-picrylhydrazyl (DPPH) were obtained from Sigma-Aldrich Co, U.S.A. Griess reagent, potassium ferricyanide, trichloroacetic acid, and ferric chloride were acquired from Ranbaxy lab, India. Carbofuran (Purity 98%) was a gift from Shetu Corporation Limited, Bangladesh. All of these chemicals were of analytical grade and were used as received.

Plant Material. Fresh leaves from *S. cymosum* were collected from the Botanical garden of Jahangirnagar University, Savar, Dhaka, Bangladesh in January 2017. Professor Nuhu Alam, a botanist at the Department of Botany, Jahangirnagar University, Bangladesh, identified and authenticated the plant. A voucher specimen (No. DACB47696) was deposited at the Bangladesh National Herbarium, for future reference.

Extract Preparation. Collected *S. cymosum* leaves were first cleaned with tap water, sun-dried, and then further dried at a moderate temperature 40 °C to make them suitable for grinding. Dried leaves were ground into fine powder by means of a blender (CM/L7360065, Jaipan, Mumbai, India). Approximately 1000 g of powdered plant materials was subjected to Soxhlet extraction according to a procedure outlined by Hossain et al.,⁵¹ using the apparatus (ZSS6203, Aldrich Soxhlet extraction apparatus, Darmstadt, Germany) at 65 °C temperature with methanol (5000 mL) as a solvent. Then the extract was filtered through Whatman 1 filter paper and dried under reduced pressure by means of rotary evaporation at a temperature of 40 ± 2 °C. Dried extracts thus obtained were kept at –20 °C for further analysis and were screened with respect to pharmacological properties; yield was 271.20 g.

Phytochemical Screening. Phytochemical screening of the methanol extracts of *S. cymosum* leaves to detect the presence of potential phytochemical constituents, including alkaloids, carbohydrates, glycosides, flavonoids, saponins, steroids, and tannins, was carried out using the method of De et al. with slight modification.⁵²

Phenolic and Flavonoid Content of *S. cymosum*. We employed HPLC coupled with a UV detector to determine phenolic acids and flavonoids in the methanol extract of *S. cymosum* leaves, according to published procedures with slight modification.^{53,54} Briefly, 1 g of the extract of *S. cymosum* leaves was dissolved in 10 mL of methanol (HPLC grade), followed by centrifugation at 6000 rpm for 10 min, and then filtration through a 0.45 μm syringe filter (Sartorius AG, Germany). The filtrate (5 mL) was passed through a 0.20 μm nylon membrane filter (Sigma, U.S.A.). An aliquot of 50 μL was diluted with 10 mL of methanol and loaded on the HPLC system (SPD-20AV, Serial no.: L20144701414AE, Shimadzu Corporation, Kyoto, Japan) equipped with a UV detector (SPD-20AV, Serial no.: L20144701414AE, Shimadzu Corporation, Kyoto, Japan). A Luna Phenomenex, C18 100A (150 × 4.60 mm, 5 μm) HPLC column was used. A linear gradient at a flow rate of 0.5 mL/min was used, and the total analytical time was approximately 35 min. A binary mobile phase consisting of solvent A (ultrapure water with 0.1% phosphoric acid)

and solvent B (pure methanol with 0.1% phosphoric acid) was applied. Elution from the column was achieved with the following gradient: 0 to 10 min of solvent B, increased from 35% to 55%; 10 to 25 min of solvent B, increased to 62%; 25 to 30 min of solvent B increased to 85%, and the final composition was kept constant up to 35 min. All solvents were of HPLC grade; detection wavelength was set at 265 nm. Identification and quantification of phenolics and flavonoids was performed by comparing the retention times of the analytes with reference standards.

Determination of the Total Phenol and Flavonoid Content, and Total Antioxidant Capacity. Total polyphenol content of the extract, expressed as g of gallic acid equivalent (GAE)/100 g of extract, was determined according to the method described by Afroz et al.⁵⁴ In a similar fashion, the total flavonoid content, given as g of quercetin equivalent (QE)/100 g of extract, was estimated by using the aluminum chloride colorimetric method described by Wang and Jiao.⁵⁵ On the other hand, the total antioxidant capacity of the leaf extract was measured following the phosphomolybdenum method outlined by Prieto and colleagues.⁵⁶ In these methods, absorbance of solutions was measured with the aid of a spectrophotometer (Shimadzu UV PC-1600, Japan) against a blank. Antioxidant capacity is displayed as g of ascorbic acid equivalent (AAE)/100 g of extract.

DPPH and NO Radical Scavenging Assays. The DPPH-free-radical-scavenging assay was carried out according to the method described by Prieto and co-workers,⁵⁶ whereas the nitric-oxide-scavenging activity was determined according to the method described by Govindarajan and colleagues.⁵⁷ In the DPPH assay, different concentrations of extract solutions (25, 50, 100, 200, 400, and 800 μg/mL) were placed into test tubes, and 2 mL of 0.004% DPPH solution in methanol was added to each test tube to make the final volume 3 mL. Then the mixture was incubated for 30 min in the dark. Absorbance of each solution was measured at 517 nm using a spectrophotometer (Shimadzu UV PC-1600, Japan) against a blank. For the NO-scavenging assay, solutions of extract (6.25, 12.5, 25, 50, 100, and 200 μg/mL) were similarly placed into test tubes, followed by the addition of 1.0 mL of 5 mM sodium nitroprusside solution to the test tubes. Thereafter, 1.2 mL of 0.5% Griess reagent was mixed with each solution, and absorbance of the mixture was measured at 517 nm. For both cases, results were expressed as percent of inhibition which was calculated using the following equation: $[(A_0 - A_1)/A_0] \times 100$, where A_0 = absorbance of the control and A_1 = absorbance of the extract or standard. IC_{50} is the concentration at which 50% of the total DPPH free radical is scavenged and can be determined by linear regression method from plotting % inhibition against corresponding concentration. Similarly, NO-scavenging activities of the extracts were expressed as the concentration of the extracts required to scavenge the NO radical by 50% (IC_{50} value), calculated using linear regression analysis by plotting the absorbance (the percentage of inhibition of NO radicals) against the concentration.

Reducing Power Capacity Assessment. The reducing power capacity of the extract was assessed according to published procedures.⁵⁸ Briefly, 2.0 mL of extract or standard of different concentrations (6.25, 12.5, 25, 50, 100, and 200 μg/mL) were placed into different test tubes, followed by the addition of 2.5 mL of 1% potassium ferricyanide solution into each test tube. Test tubes were incubated for 10 min at 50 °C to complete the reaction. Then, 2.5 mL of 10% trichloroacetic acid solution was added to each test tube, and each mixture was centrifuged at 3000 rpm for 10 min. Thereafter, 2.5 mL of supernatant solution was withdrawn from each of the mixtures and mixed with 2.5 mL of distilled water. This process was followed by the addition of 0.5 mL of 0.1% ferric chloride ($FeCl_3$) solution. Finally, absorbance of each solution was measured at 700 nm with a spectrophotometer against the blank.

Similarly, the cupric reducing antioxidant capacity of the leaf extract was performed according to the CUPRAC method described by Hussain and co-workers.⁵¹ According to this procedure, 500 μL solutions of an extract or standard of different concentrations (6.25, 12.5, 25, 50, 100, 200, 400, and 800 μg/mL) were placed into test tubes. Then 1.0 mL of 0.01 M $CuCl_2 \cdot 2H_2O$ solution, 1.0 mL of

ammonium acetate buffer (pH 7.0), 1.0 mL of 0.0075 M of neocaprin solution, and 600 μ L of distilled water were added, and the final volume of the mixture was adjusted to 4.1 mL. Mixtures were incubated for 1 h at room temperature, and the absorbance of each solution was measured at 450 nm against the blank.

Animals. Male Sprague–Dawley rats (140–180 g) were used throughout this investigation. These animals were obtained from the animal house facility of the Pharmacology Laboratory at the Department of Pharmacy, Jahangirnagar University, Savar, Dhaka, Bangladesh. Animals were given free access to standard diet and water *ad libitum* and were kept under standard conditions mentioned in the Animals By-Laws approved by the Bangladesh Association for Laboratory Animal Science. In addition, the experimental protocol was approved by the Biosafety, Biosecurity, Ethical Committee of Jahangirnagar University [Approval No.: ref no. BBEC, JU/M2018 (1)3]. This investigation was carried out at the Department of Pharmacy, Jahangirnagar University, Savar, Dhaka, Bangladesh. Finally, after acclimatization to normal laboratory conditions for 1 week, rats were used for hepatotoxicity study.

Acute Toxicity Study. According to the organization for economic cooperation and development (OECD) guideline, rats ($n = 50$) were randomly divided into five groups of ten animals each. Different doses of methanol extract (250, 500, 1000, 2000, and 4000 mg/kg body weight) were administered orally with the aid of a stainless-steel needle attached to a plastic syringe and inserted into the stomach through the esophagus. Then animals were observed for general signs of toxicity.

Experimental Design for Hematological and Hepatoprotective Study. A total of 42 rats were randomly divided into 6 groups of 7 animals each and were kept for the experimental period of 28 days, as follows:

Group I: Animals received 0.5 mL olive oil/rat with normal diet. This group served as a normal control group.

Group II: Positive control (SCL): Animals received *S. cymosum* leaves extract at a dose of 500 mg/kg BW/rat with normal diet.

Group III: Animals received CF alone at 1.5 mg/kg BW dissolved in olive oil (0.5 mL/rat) (negative control).

Group IV: Treatment group (SCL 250 + CF): Animals received *S. cymosum* extract at a dose of 250 mg/kg BW/rat and CF (1.5 mg/kg BW) dissolved in olive oil (0.5 mL/rat).

Group V: Treatment group (SCL 500 + CF): Animals received extract of *S. cymosum* at a dose of 500 mg/kg BW/rat and CF (1.5 mg/kg BW) dissolved in olive oil (0.5 mL/rat).

Group VI: Silymarin 100 + CF: Animals received silymarin at a dose of 100 mg/kg BW/rat and CF (1.5 mg/kg BW) dissolved in olive oil (0.5 mL/rat) (standard drug group).

On the other hand, doses of SCL administered to rats were based on the findings of the acute toxicity study, whereas the CF dose was selected according to the findings of Kaur et al.⁵⁹

All of the animals were treated for 28 consecutive days, and all of the oral administrations were given in the morning between 09:30 and 10:30 am. During the experiments, rats were observed daily for any unusual clinical findings and death while body weight changes were monitored on a weekly basis. We found that rats of the CF-treated groups suffered from severe trembling after the administration of CF; however, conditions improved in the third and fourth weeks of treatment except for the group treated with CF alone. After 28 days of treatment, rats in each group were deeply anesthetized with a ketamine hydrochloride injection (100 mg/kg) and sacrificed. Blood and liver tissues were collected for biochemical and histopathological examinations as indicated below.

Serum and Liver Tissue Homogenate Preparation. Blood samples (4 mL) were drawn out from the inferior vena cava by using a heparinized syringe. Blood was separated into two portions: 1 mL of blood was placed into EDTA tubes for hematological analysis, while the remaining portion was placed into plain tubes at room temperature for 30 min before centrifugation at 3000 rpm for 10 min to yield the serum needed for subsequent biochemical analysis. Liver tissues were flensed immediately from the surrounding tissues

and washed with ice-cold phosphate buffer saline, followed by weighing. Liver samples were then homogenized in phosphate buffer saline (25 mM, pH 7.4) to produce an approximately 10% (w/v) homogenate, which was centrifuged at 1700 rpm for 10 min, and the supernatant was collected prior to storage at -20°C until biochemical analysis. In addition, a portion of the liver tissues was stored in 10% formalin for histopathological examination. The relative organ weight gain of the liver was calculated by dividing the liver weight by the final body weight of each rat according to following formula: relative organ weight (%) = (wet organ weight/body weight) $\times 100$.⁴⁹

Hematological Analysis. Hematological analysis of the blood samples was carried out using established procedures with the aid of an automated Sysmex KX-21 hematology analyzer (XS 1000i, Sysmex Corporation Ltd., Japan). Parameters that were recorded included red blood cells (RBC), hemoglobin (HGB), hematocrit (HCT), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), white blood cells (WBC), lymphocytes, monocytes, neutrophil, platelets, mean platelet volume (MPV), platelet distribution width (PDW), pro-calcitonin (PCT), and platelet larger cell ratio (P-LCR).

Evaluation of Biochemical Parameters. We determined the biochemical parameters for liver function, such as serum alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase (ALP) lactate dehydrogenase (LDH) activities, triglycerides (TG), total cholesterol (TC), high density lipoprotein cholesterol (HDL-C), total bilirubin, and total protein and albumin levels by following the standard protocols of human commercial kits and by using Humalyzer 3500 (Human, Wiesbaden, Germany).

Lipid Peroxidation (LPO) Assay. Malondialdehyde (MDA), an indicator of lipid peroxidation, was determined according to the method of Ohkawa et al.⁶⁰ Briefly, 0.2 mL of tissue homogenate was mixed with 8.1% sodium dodecyl sulfate (0.2 mL), 20% acetic acid (1.5 mL), and 8% thiobarbituric acid (1.5 mL). Then, 4 mL of distilled water was added, and the mixture was heated at 95°C in a water bath for 60 min. After heating, the tubes were kept and cooled to room temperature, and the final volume was increased to 5 mL. A butanol/pyridine (15/1) mixture (5 mL) was added, and the contents were allowed to vortex for 2 min. The mixture was centrifuged at 3000 rpm for 10 min, and the absorbance of the upper organic layer was measured at 532 nm against a blank. MDA level was expressed as nmol of thiobarbituric acid reactive substances (TBARS) per mg of protein.

Determination of Antioxidant Enzymes. We employed the method of Nandi and Chatterjee for the determination of superoxide dismutase (SOD) level.⁶¹ According to this method, the liver tissue homogenates were recentrifuged at 12 000 rpm for 10 min at 4°C by means of an Eppendorf 5415D centrifuge (Hamburg, Germany), and the clear tissue supernatant was used for evaluation. Similarly, the activity of catalase (CAT) was determined in tissue homogenate using hydrogen peroxide as a substrate, as described by Aebi.⁶² On the other hand, levels of endogenous glutathione peroxidase (GPx) in the rat liver tissues were estimated following the method of Gupta and Baquer.⁶³ Levels of SOD and CAT were expressed as units/mg of protein, whereas GPx was expressed as mIU/mL.

Histopathological Evaluation. Liver tissues were fixed in 10% neutral buffered formalin for histopathological studies. Liver tissue was trimmed (5 μ m thickness) using a rotary microtome (HM 32S, Thermo Scientific, U.K.) and embedded in paraffin wax. Tissue sections were then stained with hematoxylin and eosin for histopathological investigation using established protocols, and they were photographed with the aid of an Olympus DP 72 microscope (Tokyo, Japan). Liver slides were scrutinized and the changes were distinguished and scored as none (–), mild (+), moderate (++), and severe (+++).

Statistical Analysis. Data were analyzed using SPSS (Statistical Packages for Social Science, version 20.0, IBM Corporation, New York, U.S.A.), Graphpad Prism (version 6.02), GraphPad Software Inc., San Diego, CA, U.S.A.), and Microsoft Excel 2013 (Redmond, Washington, U.S.A.). Results are expressed as the mean $\pm$ standard error of the mean (SEM). Data were subjected to one-way analysis of

variance (ANOVA), and statistical analysis was performed with the aid of Dunnett's and Tukey's multiple comparison to analyze data sets. Differences were considered significant at $p < 0.05$ and $p < 0.001$.

CONCLUSIONS

In summary, findings from this investigation suggest that leaf extract from *S. cymosum* contains alkaloids, tannins, and flavonoids. In addition, HPLC analysis indicated the presence of important antioxidants such as rutin, benzoic acid, and salicylic acid, among other things. These compounds could be responsible for the protective effect of the extract against CF-induced hepatotoxicity in Sprague–Dawley rats. Similarly, results revealed that SCL exhibits significant antioxidant activities which could be beneficial against oxidative stress and stress-related diseases and disorders. These findings may explain the medicinal use of *S. cymosum*. However, more detailed studies are required to establish the safety and efficacy of this plant.

AUTHOR INFORMATION

Corresponding Authors

*E-mail: muhammad.torequl.islam@tdtu.edu.vn.

*E-mail: mmubarak@ju.edu.jo. Tel: +962791016126. Fax: +962 6 5300253.

ORCID

Mohammad S. Mubarak: 0000-0002-9782-0835

Present Address

▲(M.S.M.) Department of Chemistry, The University of Jordan, Amman 11942, Jordan.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This research work was financially supported by the Ministry of Science and Technology, Bangladesh under Research and Development (R&D) project 2016–2017 (Reference No.: 39.012.002.005.16-218.2017/R&D/37).

REFERENCES

- (1) Hossen, M. S., Tanvir, E. M., Prince, M. B., Paul, S., Saha, M., Ali, M. Y., Gan, S. H., Khalil, M. I., and Karim, N. (2017) Protective mechanism of turmeric (*Curcuma longa*) on carb ofuran-induced hematological and hepatic toxicities in a rat model. *Pharm. Biol.* 55, 1937–1945.
- (2) Effiong, G. S., Udoh, I. E., Udo, N. M., Asuquo, E. N., Wilson, L. A., Ntukidem, I. U., and Nwoke, I. B. (2013) Assessment of hepatoprotective and antioxidant activity of *Nauclea latifolia* leaf extract against acetaminophen induced hepatotoxicity in rats. *Int. Res. J. Plant. Sci.* 4, 55–63.
- (3) Mansano, A. S., Moreira, R. A., Dornfeld, H. C., Diniz, L. G., Vieira, E. M., Daam, M. A., Rocha, O., and Selegim, M. H. (2018) Acute and chronic toxicity of diuron and carbofuran to the neotropical cladoceran *Ceriodaphnia silvestrii*. *Environ. Sci. Pollut. Res.* 25, 13335–13346.
- (4) Li, A.-N., Li, S., Zhang, Y.-J., Xu, X.-R., Chen, Y.-M., and Li, H.-B. (2014) Resources and biological activities of natural polyphenols. *Nutrients* 6, 6020–6047.
- (5) Jain, S., Dixit, V. K., Malviya, N., and Ambawatia, V. (2009) Antioxidant and hepatoprotective activity of ethanolic and aqueous extracts of *Amorphophallus campanulatus* Roxb. Tubers. *Acta Pol. Pharm.* 66, 423–428.
- (6) Pal, S., Bhattacharjee, A., Mukherjee, S., Bhattacharya, K., and Khowala, S. (2014) Antioxidant and hepatoprotective activity of

ethanolic extract of *Alocasia indica* tuber. *Am. J. Phytomed. Clin. Therap.* 2, 191–208.

(7) Morisco, F., Vitaglione, P., Amoroso, D., Russo, B., Fogliano, V., and Caporaso, N. (2008) *Mol. Aspects Med.* 29, 144–150.

(8) Lapenna, S., Gemen, R., Wollgast, J., Worth, A., Maragkoudakis, P., and Caldeira, S. (2015) Assessing herbal products with health claims. *Crit. Rev. Food Sci. Nutr.* 55, 1918–1928.

(9) Srirama, R., Santhosh Kumar, J. U., Seethapathy, G. S., Newmaster, S. G., Ragupathy, S., Ganeshaiah, K. N., Uma Shaanker, R., and Ravikanth, G. (2017) Species Adulteration in the Herbal Trade: Causes, Consequences and Mitigation. *Drug Saf.* 40, 651–661.

(10) Carvalho, A. C. B., Lana, T. N., Perfeito, J. P. S., and Silveira, D. (2018) The Brazilian market of herbal medicinal products and the impacts of the new legislation on traditional medicines. *J. Ethnopharmacol.* 212, 29–35.

(11) Amro, M. S., Teoh, S. L., Norzana, A. G., and Srijit, D. (2018) The potential role of herbal products in the treatment of Parkinson's disease. *Clin. Ter.* 169, e23–e33.

(12) Fern, K. (1997) *Plants for a future: Edible & Useful plants for a healthier world*; Permanent Publications: Hampshire, England.

(13) Dhar, K. S., Wahed, T. B., Hassan, A. H. M. N., and Wahed, S. B. (2016) Synthesis and biological evaluation of coumarin clubbed oxazines. *Int. J. Pharm. Sci. Res.* 7, 1021–1025.

(14) Hossain, M. A., and Nagooru, M. R. (2011) Biochemical profiling and total flavonoids contents of leaves crude extract of endemic medicinal plant *Corydalis terminalis* L. Kunth. *Pharmacogn. J.* 3, 25–30.

(15) Aiyegoro, O. A., and Okoh, A. I. (2010) Preliminary phytochemical screening and in vitro antioxidant activities of the aqueous extract of *Helichrysum longifolium* DC. *BMC Complement. Altern. Med.* 10, 21–28.

(16) Tanvir, E. M., Afroz, R., Chowdhury, M. A. Z., Khalil, M. I., Hossain, M. S., Rahman, M. A., Rashid, M. H., and Gan, S. H. (2015) Honey has a protective effect against chlorpyrifos-induced toxicity on lipid peroxidation, diagnostic markers and hepatic histoarchitecture. *Eur. J. Integr. Med.* 7, 525–533.

(17) Manas, D. (2014) The determination of vitamin C, total phenol and antioxidant activity of some commonly cooking spices crops used in West Bengal. *Int. J. Plant Physiol. Biochem.* 6, 66–70.

(18) Adedapo, A. A., Jimoh, F. O., Afolayan, A. J., and Masika, P. J. (2009) Antioxidant Properties of the Methanol Extracts of the Leaves and Stems of *Celtis africana*. *Rec. Nat. Prod.* 3, 23–31.

(19) Alam, M. I., and Gomes, A. (1998) Viper venom-induced inflammation and inhibition of free radical formation by pure compound (2-hydroxy-4-methoxy benzoic acid) isolated and purified from anantamul (*Hemidesmus indicus* R. BR) root extract. *Toxicol.* 36, 207–215.

(20) Saravanan, N., and Nalini, N. (2007) Inhibitory effect of *Hemidesmus indicus* and its active principle 2-hydroxy 4-methoxy benzoic acid on ethanol-induced liver injury. *Fundam. Clin. Pharmacol.* 21, 507–514.

(21) Mao, J. Q., Liu, C. M., and Yang, W. (2018) Protective effect of rutin against carbon tetrachloride-induced oxidative stress, inflammation and apoptosis in mouse kidney associated with the ceramide, MAPKs, p53 and calpain activities. *Chem.-Biol. Interact.* 286, 26–33.

(22) Domitrović, R., Jakovac, H., Vasiljev Marchesi, V., Vladimir-Knežević, S., Cvijanović, O., Tadić, Ž., Romić, Ž., and Rahelić, D. (2012) Differential hepatoprotective mechanisms of rutin and quercetin in CCl₄-intoxicated BALB/cN mice. *Acta Pharmacol. Sin.* 33, 1260–1270.

(23) Amanullah, M. M., Sekar, S., and Vincent, S. (2010) Plant growth substances in crop production: A review. *Asian J. Plant Sci.* 9, 215–222.

(24) Zhao, H., Xu, X., Na, J., Hao, L., Huang, L., Li, G., and Xu, Q. (2008) Protective effects of salicylic acid and vitamin C on sulfur dioxide-induced lipid peroxidation in mice. *Inhalation Toxicol.* 20, 865–871.

(25) Dinis-Oliveira, R. J., Sousa, C., Remiao, F., Duarte, J. A., Navarro, A. S., Bastos, M. L., and Carvalho, F. (2007) Full survival of

paraquat-exposed rats after treatment with sodium salicylate. *Free Radical Biol. Med.* 42, 1017–1028.

(26) Drew, J. E., Arthur, J. R., Farquharson, A. J., Russell, W. R., Morrice, P. C., and Duthie, G. G. (2005) Salicylic acid modulates oxidative stress and glutathione peroxidase activity in the rat colon. *Biochem. Pharmacol.* 70, 888–893.

(27) Guerrero, A., González-Correa, J. A., Arrebola, M. M., Munoz-Marin, J., Sanchez De La Cuesta, F., and De La Cruz, J. P. (2004) Antioxidant effects of a single dose of acetylsalicylic acid and salicylic acid in rat brain slices subjected to oxygen-glucose deprivation in relation with its antiplatelet effect. *Neurosci. Lett.* 358, 153–156.

(28) Cetin, D., Hacimuftuoglu, A., Tatar, A., Turkez, H., and Togar, B. (2016) The in vitro protective effect of salicylic acid against paclitaxel and cisplatin-induced neurotoxicity. *Cytotechnology* 68, 1361–1367.

(29) Crissman, J. W., Goodman, D. G., Hildebrandt, P. K., Maronpot, R. R., Prater, D. A., Riley, J. H., Seaman, W. J., and Thake, D. C. (2004) Best practices guideline: toxicologic histopathology. *Toxicol. Pathol.* 32, 126–131.

(30) Heikal, T. M., Ghanem, H. Z., and Soliman, M. S. (2011) Protective effect of green tea extracts against dimethoate induced DNA damage and oxidant/antioxidant status in male rats. *Biohealth Sci. Bull.* 3, 1–11.

(31) Nirwane, A. M., and Bapat, A. R. (2012) Effect of methanolic extract of Piper nigrum fruits in ethanol-CCl₄ induced hepatotoxicity in Wistar rats. *Der Pharmacia Lettre* 4, 795–802.

(32) Mansour, S. A., Mossa, A. T. H., and Heikal, T. M. (2009) Effects of methomyl on lipid peroxidation and antioxidant enzymes in rat erythrocytes: in vitro studies. *Toxicol. Ind. Health* 25, 557–563.

(33) Otuechere, C. A., Abarikwu, S. O., Olateju, V. I., Animashaun, A. L., and Kale, O. E. (2014) Protective effect of curcumin against the liver toxicity caused by propanil in rats. *Int. Scholarly Res. Not.* 2014, 1–8.

(34) Hii, Y. S., Lee, M. Y., and Chuah, T. S. (2007) Acute toxicity of organochlorine insecticide endosulfan and its effect on behaviour and some hematological parameters of Asian swamp eel (*Monopterus albus*, Zuiwew). *Pestic. Biochem. Physiol.* 89, 46–53.

(35) Adhikari, S., Sarkar, B., Chatterjee, A., Mahapatra, C. T., and Ayyappan, S. (2004) Effects of cypermethrin and carbofuran on certain hematological parameters and prediction of their recovery in a freshwater teleost, *Labeo rohita* (Hamilton). *Ecotoxicol. Environ. Saf.* 58, 220–226.

(36) Tung, H. T., Cook, F. W., Wyatt, R. D., and Hamilton, P. B. (1975) The anemia caused by aflatoxin. *Poult. Sci.* 54, 1962–1969.

(37) Nussey, G., Van Vuren, J. H. J., and Du Preez, H. H. (1995) Effect of copper on the haematology and osmoregulation of the Mozambique tilapia, *Oreochromis mossambicus* (Cichlidae). *Comp. Biochem. Physiol., Part C: Pharmacol., Toxicol. Endocrinol.* 111, 369–380.

(38) Celik, I., Yilmaz, Z., and Turkoglu, V. (2009) Hematotoxic and hepatotoxic effects of dichlorvos at sublethal dosages in rats. *Environ. Toxicol.* 24, 128–132.

(39) Abdel Rasoul, M., and Marei, G. (2016) Potential therapeutic effect of turmeric (*Curcuma longa*) against adverse effects of penconazole fungicide to white rats. *Int. J. Pharmacol. Toxicol.* 4, 178–186.

(40) Recknagel, R. O., Glende, E. A., Jr, Dolak, J. A., and Waller, R. L. (1989) Mechanisms of carbon tetrachloride toxicity. *Pharmacol. Ther.* 43, 139–154.

(41) Milatovic, D., Gupta, R. C., and Aschner, M. (2006) Anticholinesterase toxicity and oxidative stress. *Sci. World J.* 6, 295–310.

(42) Aberare, O. L., Okuonghae, P., Mukoro, N., Dirisu, J. O., et al. (2011) Triglycerides, total cholesterol, high density lipoprotein cholesterol and low density lipoprotein cholesterol in rats exposed to premium motor spirit fumes. *N. Am. J. Med. Sci.* 3, 277–280.

(43) Thapa, B. R., and Walia, A. (2007) Liver function tests and their interpretation. *Indian J. Pediatr.* 74, 663–671.

(44) Maxwell, E., Yusuf, O., Ihechiluru, E., and Blessing, E. (2014) Effect of *Magnifera indica* leaf extract on paracetamol-induced hepatic toxicity in rats. *Eur. J. Med. Plants* 4, 1167–1177.

(45) Soufy, H., Soliman, M., El-Manakhly, E., and Gaafa, A. (2007) Some biochemical and pathological investigations on monosex Tilapia following chronic exposure to carbofuran pesticides. *Glob. Vet.* 1, 45–52.

(46) Del Rio, D., Stewart, A. J., and Pellegrini, N. (2005) A review of recent studies on malondialdehyde as toxic molecule and biological marker of oxidative stress. *Nutr., Metab. Cardiovasc. Dis.* 15, 316–328.

(47) Rai, D. K., and Sharma, B. (2007) Carbofuran-induced oxidative stress in mammalian brain. *Mol. Biotechnol.* 37, 66–71.

(48) Piyaratne, P. S. (2018) *Extraction and Purification of (E)-Resveratrol from the Bark of Maine's Native Spruces*. Ph.D. Dissertation, University of Maine, Orono, ME.

(49) Liu, Z., Li, C., Li, M., Li, D., and Liu, K. (2004) The subchronic toxicity of hydroxysaffloryellow A of 90 days repeatedly intraperitoneal injections in rats. *Toxicology* 203, 139–143.

(50) Chandra, D., Tripathi, U. N., Srivastava, S., and Swaroop, A. (2011) Carbofuran induced biochemical toxicity in mice: protective role of Momordica charantia. *Euro. J. Exp. Bio.* 1, 106–112.

(51) Hossain, M. M., Mondal, M., Hossain, M. P., Uddin, M. S., Uddin, N., Hasan, M. M., Hossain, M. S., Mahmudul Islam, A. F. M., and Rana, M. S. (2015) In vitro antioxidant, brine shrimp lethality and antimicrobial activities of methanol and petroleum ether extracts of *Cassia renigera* fruit. *Food Measure* 9, 68–75.

(52) De, S., Dey, Y. N., and Ghosh, A. K. (2010) Phytochemical investigation and chromatographic evaluation of the different extracts of tuber of *Amorphophallus paeoniifolius* (Araceae). *Int. J. Pharm. Biol. Res.* 1, 150–157.

(53) Kumazawa, S., Hamasaka, T., and Nakayama, T. (2004) Antioxidant activity of propolis of various geographic origins. *Food Chem.* 84, 329–339.

(54) Afroz, R., Tanvir, E. M., Paul, S., Bhoomik, N. C., Gan, S. H., and Khalil, M. I. (2016) DNA damage inhibition properties of sundarban honey and its phenolic composition. *J. Food Biochem.* 40, 436–445.

(55) Wang, S. Y., and Jiao, H. (2000) Scavenging capacity of berry crops on superoxide radicals, hydrogen peroxide, hydroxyl radicals, and singlet oxygen. *J. Agric. Food Chem.* 48, 5677–5684.

(56) Prieto, P., Pineda, M., and Aguilar, M. (1999) Spectrophotometric quantitation of antioxidant capacity through the formation of a phosphomolybdenum complex: specific application to the determination of vitamin E. *Anal. Biochem.* 269, 337–341.

(57) Govindarajan, R., Rastogi, S., Vijayakumar, M., Shirwaikar, A., Rawat, A. K. S., Mehrotra, S., and Pushpangadan, P. (2003) Studies on the Antioxidant activities of *Disodium gangeticum*. *Biol. Pharm. Bull.* 26, 1424–1427.

(58) Oyaizu, M. (1986) Studies on products of browning reaction. *Eiyogaku Zasshi* 44, 307–315.

(59) Kaur, B., Khera, A., and Sandhir, R. (2012) Attenuation of cellular antioxidant defense mechanisms in kidney of rats intoxicated with carbofuran. *J. Biochem. Mol. Toxicol.* 26, 393–398.

(60) Ohkawa, H., Ohishi, N., and Yagi, K. (1979) Assay for lipid peroxides in animal tissues by thiobarbituric acid reaction. *Anal. Biochem.* 95, 351–358.

(61) Nandi, A., and Chatterjee, I. B. (1988) Assay of superoxide dismutase in animal tissues. *J. Biosci.* 13, 305–15.

(62) Aebi, H. (1984) Catalase in vitro Methods. *Methods Enzymol.* 105, 121–126.

(63) Gupta, B. L., and Baquer, N. Z. (1998) Hexokinase, glucose-6-phosphate dehydrogenase and antioxidant enzymes in diabetic reticulocytes: effects of insulin and vanadate. *IUBMB Life* 46, 1145–1152.